

Berkeley Out-of-Order Machine (BOOM) Setup

Created by **NIGIL MOHRA**

1. Setting-up Chipyard on Ubuntu

```
# Updates
sudo apt update
sudo apt install -y build-essential
sudo apt install -y git
sudo apt install -y curl
sudo apt install -y wget
sudo apt install -y autoconf
sudo apt install -y automake
sudo apt install -y libtool
sudo apt install -y pkg-config
sudo apt install -y bc
sudo apt install -y zlib1g-dev
sudo apt install -y libglib2.0-dev
sudo apt install -y libpixmap-1-dev
sudo apt install -y device-tree-compiler
sudo apt install -y ninja-build
sudo apt install -y openssh-client
```

```
# Installation of Miniconda Environment
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh
bash Miniconda3-latest-Linux-x86_64.sh
# Close and Reopen Terminal

# Check
conda config --show auto_activate

# Agree to Terms and Conditions
conda tos accept --override-channels --channel https://repo.anaconda.com/pkgs/main
conda tos accept --override-channels --channel https://repo.anaconda.com/pkgs/r

# Restart the Terminal
conda config --set auto_activate_base false

# Clone Chipyard
git clone https://github.com/ucb-bar/chipyard.git
mv chipyard Chipyard
cd Chipyard

# Updating the YAML Configuration File
nano conda-reqs/chipyard-base.yaml
```

As before: change `- sysroot_linux-64=2.43` to `- sysroot_linux-64` , and add `- python=3.10` under dependencies .

```
# Disable SPEC2026 Cloning
cd ~/Chipyard
git config -f .gitmodules submodule.software/spec2026.update none
git submodule sync

# Cloning Submodules
./scripts/init-submodules-no-riscv-tools.sh
./build-setup.sh riscv-tools -ud --use-lean-conda --skip-submodules

# Cleaning the Conda Environment
rm -rf .conda-lock-env
```

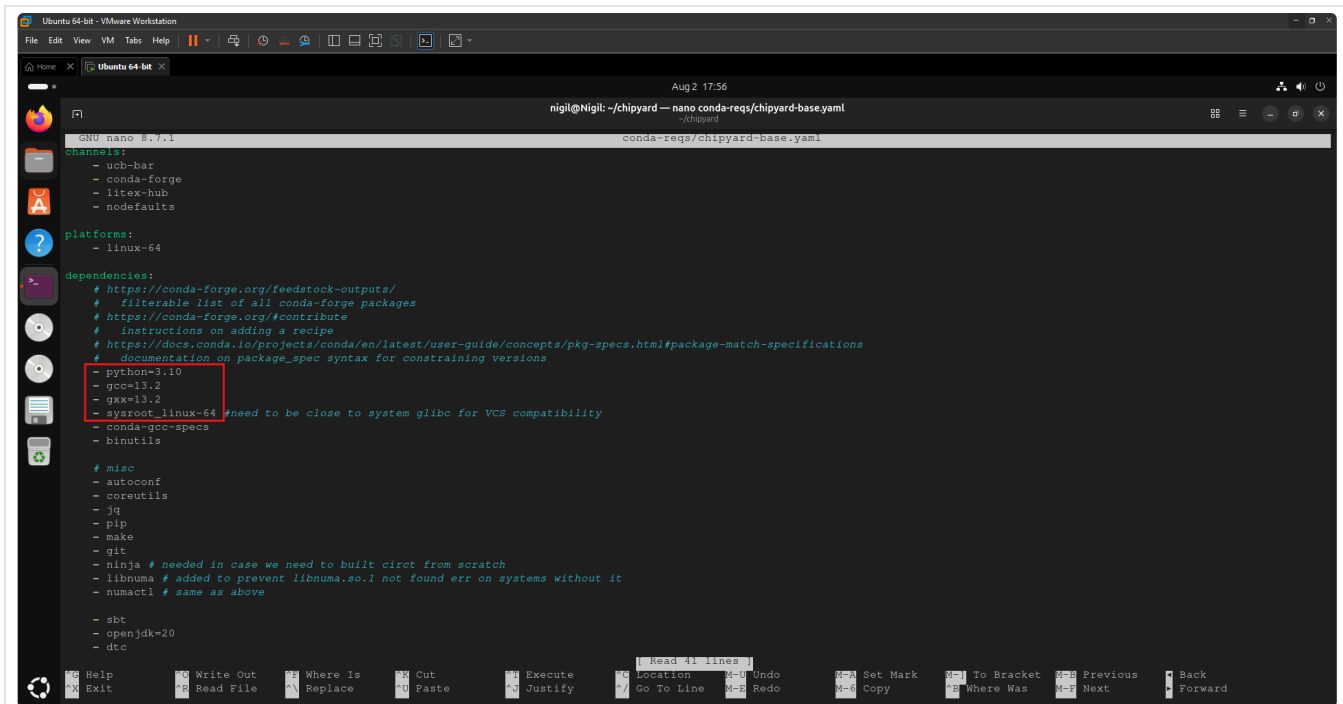


Figure 1. Updating the YAML Configuration File

```
# Install CIRCT Submodule
cd ~/Chipyard

# Invoke the Conda Environment
source ~/Chipyard/env.sh
git submodule update --init tools/install-circt

# Install
./tools/install-circt/bin/download-release-or-nightly-circt.sh \
  -f circt-full-static-linux-x64.tar.gz \
  -i "$RISCV" \
  -v version-file \
  -x conda-reqs/circt.json \
  -g unused

# Check
which firtool
firtool --version
```

The setup of the required Chipyard submodules for using the BOOM processor is now complete. The next step is to build the BOOM processor by executing the following commands.

2. Smoke Test on RV64-BOOM

```
# Configuring BOOM
cd ~/Chipyard
source ~/Chipyard/env.sh
cd sims/verilator
make CONFIG=SmallBoomV3Config debug

# Check
echo $?
ls -lh ~/Chipyard/sims/verilator/simulator-chipyard.harness-SmallBoomV3Config-debug
```

```
# Create Directory (Figure 3a)
```

```
mkdir -p ~/Chipyard/software/AESTest
```

```
# Clone Repository
```

```
cd ~/Chipyard/software/AESTest
```

```
git clone https://github.com/kokke/tiny-AES-C.git .
```

```
# Invoke Chipyard Conda Environment
```

```
source ~/Chipyard/env.sh
```

```
riscv64-unknown-elf-gcc -static -specs=htif_nano.specs -O3 aes.c test.c -o tiny_aes.riscv
```

```
# Run
```

```
cd ~/Chipyard/sims/verilator
```

```
./simulator-chipyard.harness-SmallBoomV3Config-debug \  
+permissive +max-cycles=10000000 +permissive-off \  
~/Chipyard/software/AESTest/tiny_aes.riscv
```